

UrbanStay Dataset Metadata

MIS710 Assessment 1

1. Dataset Overview

StayWise Analytics (SWA) has been engaged by UrbanStay, a fictitious Australian short-term accommodation platform, to investigate the factors influencing nightly listing prices. The dataset has been created specifically for teaching the supervised machine learning lifecycle in a business context, including data quality assessment, exploratory data analysis, multiple linear regression, model evaluation and business recommendations.

The dataset is synthetic, but designed to resemble the characteristics of a realistic business dataset. The training and challenge datasets were generated from the same synthetic population to ensure comparable data distributions while representing unseen listings.

Some variables may require preprocessing before they are suitable for machine learning. Students should justify any data preparation decisions made during their analysis. Not every available variable is necessarily an appropriate or meaningful predictor of the target variable. Students should critically evaluate the suitability of variables before including them in their predictive model.

2. Files Provided

File	Purpose
UrbanStay.csv	Labelled training dataset for the assessment 1 purpose.
UrbanStay_challenge.csv	Unlabelled dataset for the prediction challenge.
UrbanStay_metadata.docx	Dataset documentation – this document

3. Variable Dictionary

Variable	Type	Description
Listing_ID	Identifier	Unique identifier for each property listing 990003 – 998000 (integer)
Host_Name	Text	Host name (identifier rather than business feature) 998 unique names across 6,500 listings (~6–7 listings/host)

Property_Type	Categorical	Accommodation type	Apartment · House · Studio · Townhouse · Villa (5 levels)
Location	Categorical	Broad location category	CBD · Inner Suburb · Outer Suburb · Regional (4 levels)
Bedrooms	Numeric	Number of bedrooms	integer
Bathrooms	Numeric	Number of bathrooms	integer
Accommodates	Numeric	Maximum guests	1 – 10
Review_Score	Numeric	Average guest rating	average guest rating (1–5).
Number_of_Reviews	Numeric	Total reviews	integer
Superhost	Binary	Superhost status	Yes · No
Host_Experience	Numeric	Years hosting	0 – 15 (integer)
Cleaning_Fee	Numeric	Cleaning fee charged per booking (AUD).	Currency AUD
Minimum_Nights	Numeric	Minimum booking length	1 – 14 nights (integer)
Amenities_Score	Numeric	Composite score representing the range and quality of amenities available	1.0 – 10.0
Instant_Bookable	Binary	Instant booking available	Yes · No
Nightly_Price	Numeric	Nightly listing price (AUD)	Currency AUD

3. Notes for Students

The dataset represents one view of reality rather than every factor that may influence nightly prices. Some relevant business factors are unavailable, reflecting real-world analytics projects where complete information is rarely accessible.

Like most real-world datasets, this dataset provides useful evidence rather than complete knowledge. Your task is to develop the most appropriate model from the available evidence while recognising its limitations.